Laboratory 06

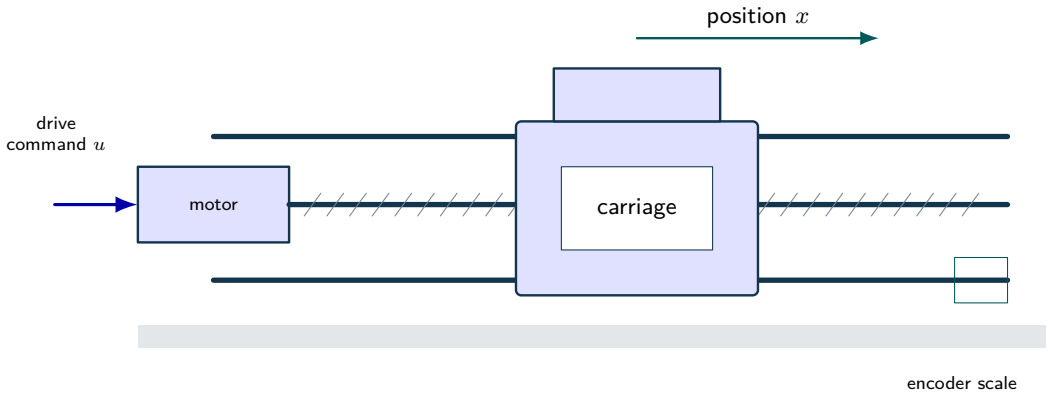
Gain selection with a third-order actuator

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal unity negative-feedback loop: $G(s) = 1/[s(s+2)(s+6)]$ and scalar controller K. The positioning-axis picture illustrates a possible application; the normalized coefficients are stipulated.

1. A position actuator with an identified aggregate model

The sketch shows a possible motor-driven carriage. Its normalized plant $1/[s(s+2)(s+6)]$ is stipulated as an aggregate teaching model; the poles are not inferred from the drawing dimensions.



Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

A motorized positioning axis illustrates the role of the integrator and two stable lags. The normalized transfer function is specified as a teaching model; the pictured dimensions do not identify its coefficients.

(a) Consider unity negative feedback with loop $L(s) = K/[s(s+2)(s+6)]$, $K > 0$. Derive the reference transfer function, characteristic polynomial and open-loop poles and zeros.

(b) Construct the cubic Routh first column. Find the full strictly stable positive-gain interval and classify its endpoints.

(c) A numerical step plot appears bounded for $K = 97$ over a short interval. Use the algebraic result to decide whether this is sufficient evidence of stability and specify a decisive numerical check.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. Following branches through a breakaway

Use the open-loop pole map to construct the locus geometrically, then verify the repeated-pole event algebraically. Reject stationary points that belong to the wrong gain sign.

(a) Determine the positive-gain real-axis root-locus segments, asymptote centroid and asymptote angles. Explain how many branches reach infinity.

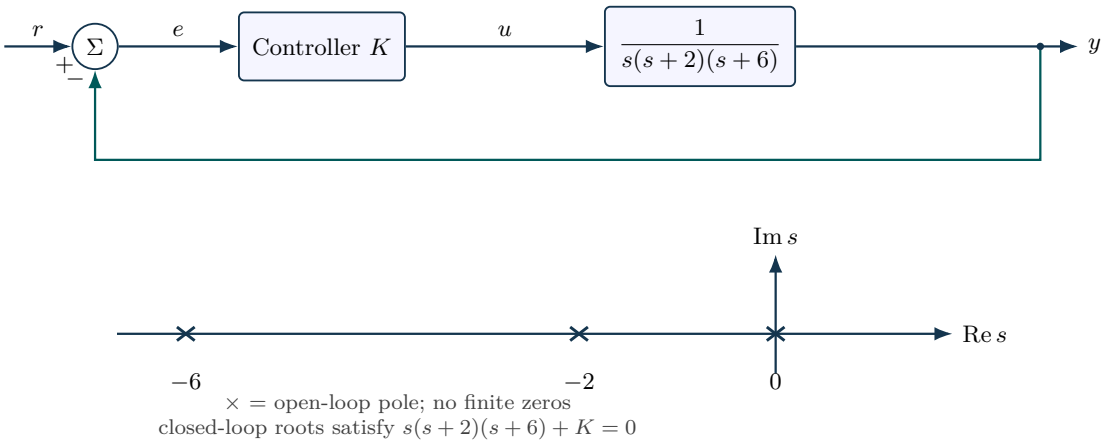
(b) Find the stationary points of $K(s) = -s(s+2)(s+6)$ on the real axis. Select the positive-gain breakaway point and calculate its gain.

(c) At the valid breakaway, determine the third pole without solving another cubic. Check the repeated-pole factorization against the coefficient of s^2.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. Boundary oscillations and feedback-sign mistakes

Classify the nominal loop at both stability boundaries and after reversing the controller sign. Distinguish input–output cancellation from decay of internal states.



Unity negative feedback and the open-loop pole map. Pole symbols provide the starting geometry for the root-locus exercises.

(a) At the upper stability boundary, calculate the oscillation frequency and period and the third pole. Explain whether a bounded-looking oscillation over a finite simulation interval establishes asymptotic stability.

(b) For $K < 0$, prove that at least one real closed-loop pole is positive without using a root solver. Relate the result to the feedback sign.

(c) At $K = 0$, the reference-to-output transfer has zero numerator. Does that imply every internal state converges to zero? Explain using the unreduced plant modes and nonzero initial conditions.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. A tracking specification with gain uncertainty

Proportional gain must satisfy both a ramp-error specification and strict stability. Treat these inequalities jointly rather than increasing gain until the plot looks acceptable.

(a) For stable K, derive the position and velocity error constants. Find the unit-step and unit-ramp steady errors at $K = 24$. State the stability qualification.

(b) Require unit-ramp error no larger than 0.1 using only positive proportional K. Combine the error and stability conditions. What design conclusion follows?

(c) Relax the allowed unit-ramp error to 0.15 and select nominal $K = 84$. If actual gain is $84(1 + \delta)$ with $|\delta| \leq \varepsilon$, find the largest symmetric uncertainty bound that preserves both requirements.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. Changing time scale and canceling a stable mode

A redesign changes actuator time scale and adds a compensator. Keep the physical internal mode visible when reducing a transfer function.

(a) Replace the plant denominator by $s(s + 2a)(s + 6a)$ with $a > 0$. Derive the stable gain interval and boundary frequency, and evaluate both at $a = 0.5$.

(b) Suppose the controller is $C = K(s+2)/(s+10)$. Under exact cancellation, derive the reduced characteristic cubic and its stability interval. Explain why the physical realization still contains a stable internal mode and why mismatch matters.

(c) Combine the changes: use plant $1/[s(s + 2a)(s + 6a)]$ and controller $K(s + 2a)/(s + 10a)$. Derive the reduced stability interval, boundary frequency and hidden stable pole; evaluate them for $a = 0.5$.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.